

# E-Commerce Retail Sales Analysis

## SQL Data Cleaning and Analysis Documentation

### Project Overview

This project analyzes e-commerce retail sales data using SQL and Power BI. The objective was to clean, validate, and transform raw transactional data into a reliable dataset for business reporting and dashboard development.

### Dataset Information

- Dataset: E-Commerce Retail Sales Dataset
- Total Records: 541,909
- Analysis Period: 2010–2011
- Tools Used:
  - PostgreSQL
  - SQL
  - Power BI
  - Power Query

### Data Cleaning Process

#### 1. Data Exploration

Initial exploration was conducted to understand the dataset structure, identify data types, and assess data quality issues.

Key checks performed:

- Total number of records
- Missing values
- Duplicate records
- Invalid quantities
- Invalid prices

#### 2. Data Quality Assessment

The following issues were identified:

| Issue | Count |
|-------|-------|
|-------|-------|

|                           |         |
|---------------------------|---------|
| Missing Customer IDs      | 135,080 |
| Missing Descriptions      | 1,454   |
| Negative Quantities       | 10,624  |
| Zero/Negative Unit Prices | 2,517   |

### 3. Data Transformation

Several transformations were applied to prepare the dataset for analysis:

- Converted InvoiceDate to proper date format.
- Extracted Month Name and Month Number.
- Created Revenue column (Quantity × UnitPrice).
- Standardized data types.
- Removed or flagged invalid records where necessary.

### Business Questions Explored

1. What is the total revenue generated?
2. How many orders were processed?
3. How many unique customers made purchases?
4. Which countries generated the highest revenue?
5. What are the monthly sales trends?
6. Which products generated the most revenue?

### Key Findings

- Total Revenue: \$9.75 Million
- Total Orders: 541.9K
- Customers: 4K
- Countries Served: 38
- The United Kingdom generated the highest revenue.
- November recorded the highest monthly sales performance.
- Revenue increased significantly in Quarter 4.

### Dashboard Development

The cleaned dataset was imported into Power BI to create an interactive dashboard featuring:

- Revenue KPI
- Total Orders KPI
- Customer Count KPI
- Countries Served KPI

- Revenue by Quarter
- Monthly Sales Trend
- Top Countries by Revenue
- Top Products by Revenue

## **Conclusion**

The project demonstrates the end-to-end analytics workflow, including data cleaning, SQL analysis, data quality assessment, and dashboard development. The resulting dashboard provides actionable insights into sales performance, customer activity, and revenue trends across multiple countries.